F839 — the mapping derivation (my piece, K1219), and it lands on the honest NEGATIVE branch: at the forced lepton positions {5/2, 3/2, 0} (T2517), the position set is NOT Weyl-symmetric, so the Gram-covariance isotropy that Koide requires is NOT forced — and my derivation SUPPORTS Elie's $A^2 = 1.19$, not 2. Keeper's re-ordering is right: Weyl-invariance is a property of the kernel, but the covariance depends on WHERE the modes sit, and the modes only give a round (isotropic) covariance if the position set is itself Weyl-symmetric. It isn't. From T2517: electron at $\nu = 5/2 = \rho_1$ (the self-shadow fixed point of $\nu \to 5-\nu$), muon at $\nu = 3/2 = \rho_2$ (Cartan slice), tau at $\nu = 0$ (Shilov). Symmetry test: (i) the $e_1 \leftrightarrow e_2$ swap sends (5/2 on axis-1, 3/2 on axis-2) → (5/2 on axis-2, 3/2 on axis-1) ≠ original (would need 5/2 = 3/2) — BROKEN; (ii) the reflection $\nu \to 5-\nu$ sends {5/2, 3/2, 0} → {5/2, 7/2, 5} — only the electron is fixed, muon and tau reflect OUT of the lepton set — BROKEN. So the diagonal covariance entries go as ${\rho_1}^2$ vs ${\rho_2}^2 = (5/2)^2$ vs $(3/2)^2 = 6.25$ vs 2.25 (ratio 2.78, not 1) → ANISOTROPIC → $A^2 \neq$ rank, consistent with Elie's 1.19. THE TENSION, stated flat: "Koide = isotropy of the $\sqrt{}$mass covariance" is an EXACT identity (F836), but the FK/Bergman mechanism at the forced positions does NOT produce that isotropy — so the mechanism does not reproduce Koide's exact $A^2 = 2$ at {5/2,3/2,0}. Either the Koide-from-FK-bulk mechanism has a real gap, OR a missing ingredient (the proper genus-5 kernel weighting, or a mode placement different from my naive ρ-axis assignment) restores isotropy — that last is the one honest opening, and it's Grace's independent mapping + Elie's proper kernel to decide. My derivation leans NEGATIVE and I report it straight, not nudged toward 2.

Lyra, Thursday 2026-08-06. My assigned derivation — and it came out against the pretty mechanism I found yesterday (F837). That's the discipline working on my own result: I built the isotropy story, and my careful check of the mapping says the forced positions break it. Calibrate both directions: I don't soften the negative to save F837, and I hold the one genuine opening (the exact mode placement is Grace's to pin, and might not be my naive ρ-axis assignment).

The mapping (forced, from T2517 — reconnect, not invent)

- electron: ν = n_C/2 = 5/2 = ρ_1 — the self-shadow fixed point of $\nu \to$(n_C−ν)=5−ν, pinned three independent ways (T2517).
- muon: ν = N_c/2 = 3/2 = ρ_2 — Wallach degeneration, the Cartan slice.
- tau: $\nu = 0$ — Wallach degeneration, the Shilov points. So the three modes sit at the two ρ-components {5/2, 3/2} plus 0.

The symmetry test (the decisive step — my piece)

Isotropy (A^2 = rank) requires the covariance to be round, which the B_2 Weyl group (order 8, the 90° rotation) forces ONLY IF the position set is Weyl-invariant. It is not: - $e_1 \leftrightarrow e_2$ swap: (5/2 on axis-1, 3/2 on axis-2) → (5/2 on axis-2, 3/2 on axis-1) ≠ original. Would need 5/2 = 3/2. BROKEN. - $\nu \to 5-\nu$ reflection: {5/2, 3/2, 0} → {5/2, 7/2, 5}; only the electron (5/2) is fixed; muon (3/2→7/2) and tau (0→5) reflect OUT of the lepton set. BROKEN. - Consequence: diagonal covariance ~ ρ_1^2 vs ρ_2^2 = 6.25 vs 2.25 (ratio 2.78) → ANISOTROPIC → $A^2 \neq$ rank. Consistent with Elie's $A^2 = 1.19$.

The honest conclusion (the negative branch)

"Koide ⍰ isotropy of the $\sqrt{}$mass covariance" is an EXACT identity (F836). But the FK/Bergman mechanism at the forced positions does NOT produce that isotropy — the positions break the symmetry that would force it. So the FK-bulk mechanism, as it stands, does not reproduce Koide's exact $A^2 = 2$ at {5/2,3/2,0}; my derivation supports $A^2 = 1.19$ (the honest branch, not a proxy artifact).

The one genuine opening (Grace + Elie decide)

My symmetry test uses the NAIVE placement (electron on the e_1/ρ_1 axis, muon on the e_2/ρ_2 axis). The exact mode→polydisk mapping is Grace's independent piece — if the modes place differently (not my naive ρ-axis assignment), or if the proper genus-5 kernel weighting restores roundness despite asymmetric positions, isotropy could survive. I hold that open. But the burden is now on finding that ingredient; the default (naive placement + proxy) is anisotropic, and the forced positions are genuinely asymmetric.

Tier / handoffs

- @Grace — your independent mapping is the arbiter of the branch I'm leaning negative on: does {5/2,3/2,0} place onto the polydisk in a way that's Weyl-symmetric (→ isotropic, A^2=rank) or not (→ 1.19)? My naive ρ-axis placement says NOT symmetric (5/2≠3/2 breaks the swap; μ,τ reflect out of the set under $\nu \to 5-\nu$). If your independent mapping agrees, the isotropy mechanism has a real gap. If it finds a symmetric placement I missed, that's the save — show it.
- @Elie — the proper genus-5 Gram matrix is the other arbiter: does the kernel weighting restore roundness despite the asymmetric positions, or confirm your 1.19? If 1.19 stands, that's the honest answer — report it; don't tune the kernel or the mapping to reach 2.
- @Cal — my mapping derivation SUPPORTS the negative branch (positions not Weyl-symmetric → $A^2 \neq$ rank). Guard the pre-registration: A^2 = rank must come from a symmetric placement (Grace) computed forward, not from choosing a kernel/mapping to reach 2. And note I'm reporting AGAINST my own F837 mechanism — the isotropy identity is exact, but the FK mechanism producing it at these positions is now doubtful.
- @Keeper — my piece (the mapping) lands negative: the forced positions {5/2,3/2,0} break both the $e_1 \leftrightarrow e_2$ swap and the $\nu \to 5-\nu$ reflection, so isotropy is NOT forced and A^2=1.19 is the likely real answer — the Koide=FK-bulk-isotropy mechanism has a genuine gap at the forced positions. The one opening (a non-naive placement or kernel weighting) is Grace's + Elie's to find. Koide PRL stays Conditional-Forced, and the honest lean is now toward "not forced at these positions." Vindicates the fork we flagged.
- @Casey — I derived the piece you'd want me to, and it went against the pretty answer I gave you yesterday — which is the honest outcome, so here it is straight. Yesterday I said Koide's two-thirds is the $\sqrt{}$masses being "perfectly round" on your two torus-handles. Today I checked WHERE the three leptons actually sit on those handles — the positions were forced a month ago — and they're lopsided: the electron sits far out on one handle (5/2), the muon closer in on the other (3/2), the tau at the center. Lopsided positions can't give a round answer — and that's exactly the off-round 1.19 Elie measured, not a computational glitch. So my own careful check says the beautiful mechanism probably doesn't hold at the positions the geometry actually forces. There's one way it could still survive — if the deeper mode-shapes place more symmetrically than their bare positions suggest, or the proper overlap weighting rounds them out — and Grace is checking that independently, precisely so I'm not both the one who proposed it and the one who judges it. But I'd

rather tell you tonight that the odds just moved *against* my own idea than let it ride on elegance. If it dies, we learned the lepton masses are genuinely lopsided and Koide's roundness comes from something subtler; if Grace finds the symmetric placement I missed, it lives. Either way, honest. Nothing pushed.

Notes only; no toy/theorem claimed (mapping derivation, K1219; leans negative). F839: forced lepton positions {5/2,3/2,0} (T2517: e=5/2=ρ_1 self-shadow fixed pt, μ=3/2=ρ_2 Cartan slice, τ=0 Shilov). SYMMETRY TEST: $e_1 \leftrightarrow e_2$ swap BROKEN (5/2≠3/2); $\nu \to 5-\nu$ BROKEN ({5/2,3/2,0}→{5/2,7/2,5}, only e fixed, μ/τ reflect out). → covariance ANISOTROPIC (diag $\rho_1{}^2$:$\rho_2{}^2$=6.25:2.25, ratio 2.78) → A^2≠rank, consistent with Elie 1.19. "Koide=isotropy" is exact (F836) but the FK mechanism at forced positions does NOT produce it → mechanism has a GAP; A^2=1.19 likely real, NOT a proxy artifact. ONE OPENING: non-naive mode placement (Grace's mapping) or genus-5 kernel weighting (Elie) restores isotropy. My derivation leans NEGATIVE, reported against my own F837. @Grace independent mapping = arbiter; @Elie proper Gram = arbiter, 1.19 = honest answer; @Cal A^2=rank must be from symmetric placement forward, not nudged. Koide PRL Conditional-Forced, lean now negative. — Lyra